

Military Institute of Science and Technology

Department of Computer Science & Engineering

Undergraduate Thesis

**AI Enhanced Complaint Management
System for Government Marketplaces**

Submitted by

Syed Mafijul Islam

Nahidur Zaman

Sakif Shahrear

Supervised by

M. Aktaruzzaman

Associate Professor and Head

Department of Robotics and Mechatronics Engineering

Daffodil International University

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Engineering

May 2026

Declaration

We hereby declare that this thesis titled *AI Enhanced Complaint Management System for Government Marketplaces* has been completed as part of the undergraduate research requirement. The work presented in this thesis is based on our own design, implementation, experimentation and analysis, except where external sources have been cited using standard academic conventions.

We further declare that the dataset, experimental results and machine learning performance values reported in this thesis are not manually invented. The reported evaluation values are obtained from the implemented training pipeline using the 500-complaint dataset exported from the NityMulya backend. Any limitation related to dataset size, backend feature dependence, class imbalance or model generalization has been discussed honestly in the relevant chapters.

Syed Mafijul Islam

Date

Nahidur Zaman

Date

Sakif Shahrear

Date

Approval Sheet

This thesis titled *AI Enhanced Complaint Management System for Government Marketplaces*, submitted by Syed Mafijul Islam, Nahidur Zaman and Sakif Shahrear, has been accepted as satisfactory in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

M. Aktaruzzaman

Date

Supervisor

Associate Professor and Head

Department of Robotics and Mechatronics Engineering

Daffodil International University

Brigadier General Mohammad Shahjahan Majib

Date

Head of Department Department of Computer Science & Engineering

Military Institute of Science and Technology

Acknowledgement

We express our sincere gratitude to Almighty Allah for giving us the strength, patience and opportunity to complete this undergraduate thesis. We are deeply thankful to our supervisor, M. Aktaruzzaman, for his guidance, encouragement and academic direction throughout the research work.

We also acknowledge the support of our department, teachers and classmates. Their comments helped us refine the research problem and improve the presentation of the work. Discussions on software engineering, database design, Bengali language processing and responsible machine learning shaped many parts of this study.

Finally, we thank the people and practical setting behind the NityMulya marketplace platform. Their context gave this work its direction. Our aim was simple: to design a lightweight complaint classification pipeline that can support government marketplace administrators without expensive infrastructure or very large language models.

Abstract

Digital government marketplaces depend heavily on citizen feedback to monitor service quality and identify product anomalies. In Bangladesh, these complaints are often written informally using mixed scripts, shorthand, and spelling variations across Bengali, English, and “Banglish.” Manual triage is slow and inconsistent, while large transformer models remain too expensive for public service systems with limited computing resources.

This work introduces a lightweight, hybrid machine learning pipeline for the NityMulya platform using a dataset of 500 complaints stored in a Supabase PostgreSQL backend. The pipeline cleans text using Unicode normalization, punctuation cleanup, and a custom Banglish dictionary. It then fuses word and character-level TF-IDF text features with numerical backend features (validity and priority scores). We evaluated multiple classifiers using stratified five-fold cross-validation, applying SMOTE strictly inside training loops to prevent data leakage.

Our final tuned Logistic Regression classifier achieved a hold-out accuracy of 86.00% and a weighted F1-score of 0.8594 on a 20% test split. An ablation study showed that text-only features yielded a weak F1-score of 0.5559, but combining text with backend validity and priority attributes dramatically boosted cross-validation performance to 0.8503. Product category features did not improve the signal. This research contributes a practical, CPU-friendly, and cost-effective decision-support tool tailored for low-resource e-governance complaint triage.

Keywords: Bengali NLP, Banglish normalization, complaint classification, e-governance, TF-IDF, Logistic Regression, Supabase, feature fusion.

সারসংক্ষেপ

সরকারি ডিজিটাল বাজারগুলোতে সেবার মান তদারকি এবং পণ্যের অসঙ্গতি সনাক্ত করতে নাগরিকদের দেওয়া অভিযোগের উপর ব্যাপকভাবে নির্ভর করতে হয়। বাংলাদেশে এই অভিযোগগুলো সাধারণত বাংলা, ইংরেজি এবং মিশ্র স্ক্রিপ্ট বা ”বাংলিশ” ব্যবহার করে অনানুষ্ঠানিক উপায়ে লেখা হয়, যার মধ্যে প্রচুর বানানভেদ ও সংক্ষিপ্ত রূপ দেখা যায়। এই অভিযোগগুলো হাতে কলমে বাছাই করা অত্যন্ত ধীরগতির এবং অসামঞ্জস্যপূর্ণ, অন্যদিকে সীমিত কম্পিউটিং রিসোর্স সম্পন্ন সরকারি সেবা ব্যবস্থার জন্য বড় বড় ট্রান্সফর্মার মডেল ব্যবহার করা বেশ ব্যয়বহুল।

এই গবেষণায় Supabase (PostgreSQL) ব্যাকএন্ডে সংরক্ষিত 500টি অভিযোগের একটি ডেটাসেট ব্যবহার করে ‘নিত্যমূল্য’ প্ল্যাটফর্মের জন্য একটি হালকা ও হাইব্রিড মেশিন লার্নিং পাইপলাইন তৈরি করা হয়েছে। পাইপলাইনটি ইউনিকোড নরমালাইজেশন, বিরামচিহ্ন পরিষ্কার এবং একটি কাস্টম বাংলা শিথি অভিধান ব্যবহার করে টেক্সট ডেটা প্রক্রিয়াজাত করে। এরপর এটি শব্দ এবং ক্যারেক্টার-লেভেলের TF-IDF টেক্সট ফিচারের সাথে ব্যাকএন্ডের গাণিতিক ফিচারসমূহ (বৈধতা এবং অগ্রাধিকার স্কোর) একত্রিত করে। আমরা ডেটা লিকেজ রোধ করতে কঠোরভাবে ট্রেনিং লুপের ভেতরে SMOTE প্রযুক্তি প্রয়োগ করে স্ট্র্যাটিফাইড ফাইভ-ফোল্ড ক্রস-ভ্যালিডেশনের মাধ্যমে বেশ কয়েকটি ক্লাসিফায়ার মূল্যায়ন করেছি।

আমাদের চূড়ান্ত টিউনড লজিস্টিক রিগ্রেশন ক্লাসিফায়ারটি 20% টেস্ট স্প্লিটে 86.00% অ্যাকুরেসি এবং 0.8594 ওয়েটেড F1-স্কোর অর্জন করেছে। একটি অ্যাবলেশন স্টাডিতে দেখা গেছে যে, শুধুমাত্র টেক্সট ফিচারের F1-স্কোর ছিল মাত্র 0.5559, কিন্তু টেক্সটের সাথে ব্যাকএন্ডের বৈধতা ও অগ্রাধিকার ফিচার যুক্ত করায় ক্রস-ভ্যালিডেশন পারফরম্যান্স এক লাফে 0.8503-এ উন্নীত হয়। পণ্যের ক্যাটাগরি ফিচারটি মডেলের কার্যকারিতা বাড়াতে কোনো ভূমিকা রাখেনি। এই গবেষণাটি সীমিত সম্পদের ই-গভর্নেন্স অভিযোগ ট্রিয়াজের জন্য একটি বাস্তবসম্মত, CPU-friendly এবং সাশ্রয়ী সিদ্ধান্ত-সহায়ক টুল তৈরিতে অবদান রাখে।

মূলশব্দ: বাংলা এনএলপি, বাংলা নরমালাইজেশন, অভিযোগ শ্রেণীকরণ, ই-গভর্নেন্স, TF-IDF, লজিস্টিক রিগ্রেশন, সুপাবেস, ফিচার ফিউশন।

Contents

Declaration	ii
Approval Sheet	iii
Acknowledgement	iv
Abstract	v
সারসংক্ষেপ	vi
Abbreviation List	xv
1 Introduction	1
1.1 Background of E-Governance Systems	1
1.2 Government Marketplace Complaint Management	2
1.3 Problems in Manual Complaint Processing	2
1.4 Bengali and Banglish NLP Challenges	3
1.5 Motivation of the Study	3
1.6 Problem Statement	3
1.7 Research Objectives	4
1.8 Research Questions	4
1.9 Scope of the Study	5
1.10 Research Contribution	5
1.11 Thesis Organization	6
2 Literature Review	7
2.1 Introduction to Natural Language Processing	7

2.2	Text Classification Systems	7
2.3	Bengali NLP Research	8
2.4	Banglish Language Processing Challenges	8
2.5	TF-IDF Feature Extraction	9
2.6	Character N-Grams	9
2.7	Logistic Regression for NLP	10
2.8	Support Vector Machines for Text Classification	10
2.9	Random Forest for Classification	10
2.10	SMOTE and Imbalance Handling	11
2.11	E-Governance Complaint Systems	11
2.12	Existing Bengali Complaint Classification Works	11
2.13	Research Gaps	12
3	System Overview and Dataset	14
3.1	NityMulya Platform Overview	14
3.2	System Workflow	14
3.3	Supabase Backend Architecture	15
3.4	PostgreSQL Storage Design	15
3.5	Complaint Submission Process	16
3.6	Dataset Collection Pipeline	16
3.7	Dataset Characteristics	17
3.8	Data Distribution	18
3.9	Label Definition	18
3.10	Complaint Examples	19
3.11	Ethical Considerations	19
3.12	Data Privacy Discussion	20
4	Methodology	21
4.1	Overall Machine Learning Pipeline	21
4.2	Text Preprocessing Pipeline	22
4.3	Unicode Normalization	22
4.4	Banglish Normalization	22

4.5	Stopword Removal	23
4.6	TF-IDF Mathematics	23
4.7	Word N-Gram Extraction	23
4.8	Character N-Gram Extraction	23
4.9	FeatureUnion Explanation	24
4.10	Numeric Feature Scaling	24
4.11	One-Hot Encoding	24
4.12	Cross Validation Methodology	24
4.13	Hyperparameter Tuning	25
4.14	Class Imbalance Handling	25
4.15	Ablation Study Methodology	25
4.16	Why Linear Models Outperform Random Forest on Sparse TF-IDF .	26
4.17	Pseudocode	26
5	Implementation	27
5.1	Development Environment	27
5.2	Python Ecosystem	27
5.3	Scikit-Learn	27
5.4	Imbalanced-Learn	28
5.5	Supabase Integration	28
5.6	Data Loading	28
5.7	Pipeline Implementation	29
5.8	Model Training Implementation	29
5.9	Evaluation Implementation	29
5.10	Visualization Generation	30
5.11	Training Workflow Diagram	30
5.12	Deployment Possibilities	30
6	Results and Analysis	31
6.1	Evaluation Metrics	31
6.2	Baseline Random Forest Results	31
6.3	Improved Logistic Regression Results	32

6.4	TF-IDF Comparison Analysis	32
6.5	Model Comparison Analysis	33
6.6	Cross Validation Analysis	33
6.7	SMOTE Analysis	34
6.8	Ablation Study Analysis	34
6.9	Why Backend Numeric Features Improved Results	35
6.10	Why Product Category Features Reduced Performance	35
6.11	Confusion Matrix Analysis	35
6.12	Feature Importance Discussion	36
6.13	Error Analysis	37
6.14	Class-Wise Analysis	37
6.15	Statistical Interpretation	37
6.16	SVD and Cluster Visualization	37
7	Discussion	39
7.1	Interpretation of Results	39
7.2	Why the Hybrid System Works	39
7.3	Practical Deployment Implications	39
7.4	Lightweight ML Advantages	40
7.5	CPU Friendly Deployment Discussion	40
7.6	Risks of Backend Feature Dependency	40
7.7	Generalization Concerns	40
7.8	Real-World Applicability	41
7.9	Comparison with Transformer-Based Systems	41
7.10	Reviewer Concern: Data Leakage	41
7.11	Reviewer Concern: Backend Features Encoding Labels	41
7.12	Reviewer Concern: Small Dataset Size	42
8	Limitations and Future Work	43
8.1	Small Dataset Limitations	43
8.2	Class Imbalance	43
8.3	Backend-Rule Dependency	43

<i>CONTENTS</i>	xi
8.4 Banglish Normalization Limitations	44
8.5 Lack of Transformer Benchmarking	44
8.6 Generalization Limitations	44
8.7 Future Work	44
9 Conclusion	45
9.1 Summary of the Work	45
9.2 Summary of Findings	45
9.3 Contributions	45
9.4 Practical Significance	46
9.5 Final Remarks	46
10 Engineering Considerations, Challenges, and Remedies	47
10.1 Design Constraints	47
10.1.1 Ethical Considerations	47
10.1.2 Environmental Considerations	47
10.1.3 Social and Sustainability Considerations	48
10.1.4 Economic and Budgetary Constraints	48
10.2 Complex Engineering Problem (CEP)	48
10.3 Complex Engineering Activities (CEA)	49
10.4 Mapping and Justification of Program Outcomes	50
10.4.1 PO6: The Engineer and Society	50
10.4.2 PO7: Environment and Sustainability	51
10.4.3 PO8: Ethics	51
A Sample Dataset Entries	52
B Banglish Normalization Dictionary	53
C Hyperparameter Search Space	54
D Full Classification Reports	55
E Source Code Snippets	56

F Supabase Schema Examples**58**

List of Figures

3.1	High-level workflow of the NityMulya complaint classification system	15
3.2	Simplified database schema used by the complaint dataset	16
3.3	Dataset collection and experimental preparation flow	17
3.4	Distribution of complaint classes in the 20% stratified hold-out test set	18
4.1	Machine learning pipeline used in the thesis	21
5.1	Training and evaluation workflow	30
6.1	Confusion matrix for the final tuned Logistic Regression model	36
6.2	Two-dimensional SVD projection of hybrid complaint features	38
6.3	K-Means cluster projection over transformed hybrid features	38

List of Tables

2.1	Comparison with representative related work	12
3.1	Dataset class distribution	17
3.2	Backend numeric feature summary	18
3.3	Representative complaint examples	19
6.1	Final hold-out classification report for tuned Logistic Regression . . .	32
6.2	Five fold TF-IDF comparison using Linear SVC	32
6.3	Five fold model comparison on the full hybrid feature set	33
6.4	Ablation study with tuned Logistic Regression	34
10.1	Complex Engineering Problem (CEP) Justification	49
10.2	Complex Engineering Activities (CEA) Justification	50
A.1	Sample anonymized complaint records	52
B.1	Selected Banglish normalization entries	53
C.1	Hyperparameter search space	54
D.1	Confusion matrix rows=true, columns=predicted	55
D.2	Hyperparameter tuning results	55

Abbreviation List

Abbreviation	Meaning
API	Application Programming Interface
CV	Cross Validation
DBMS	Database Management System
F1	Harmonic mean of precision and recall
FN	False Negative
FP	False Positive
ICT	Information and Communication Technology
ML	Machine Learning
NFKC	Normalization Form Compatibility Composition
NLP	Natural Language Processing
OHE	One-Hot Encoding
RDBMS	Relational Database Management System
RF	Random Forest
SMOTE	Synthetic Minority Oversampling Technique
SQL	Structured Query Language
SVC	Support Vector Classification
TF-IDF	Term Frequency-Inverse Document Frequency
TN	True Negative
TP	True Positive
UI	User Interface

Chapter 1

Introduction

1.1 Background of E-Governance Systems

E-governance means using information and communication technologies to improve public services. Governments now use digital platforms for citizen registration, subsidy management, online payment, licensing, grievance redress and marketplace monitoring. Citizens do more than read information on these platforms. They submit feedback, complaints, documents and evidence. Over time, the platform becomes a steady source of operational data.

Complaint management is a central part of e-governance because it links citizens with administrative action. A complaint may report overpricing, damaged goods, expired products, misleading information, rude shop behavior or consumer rights violations. Fast handling can build public trust and help administrators find repeated problems. Slow or inconsistent handling has the opposite effect. Citizens may start to see the platform as a form-submission tool, not as a responsive public service.

The NityMulya government marketplace platform is a suitable setting for this research. It supports marketplace information, price awareness and complaint submission for essential products. Citizens write in everyday language, not in a fixed administrative format. That gives us useful real-world data but it also creates a technical challenge. The complaint text is often short, informal, mixed-language and noisy.

1.2 Government Marketplace Complaint Management

Government marketplace complaint management is different from ordinary customer service ticketing. A commercial help desk usually handles issues between a company and its customers. A government marketplace complaint may involve citizens, shop owners, product categories, legal price ranges, public-health concerns and administrative enforcement. A complaint about expired milk, false weight or price manipulation is not only a service issue. It may require regulatory attention.

The usual workflow includes complaint submission, initial review, severity assessment, routing to the responsible authority, investigation, response and archival. In a manual workflow, administrators read each complaint and decide whether it is weak, medium or high priority. The decision may depend on the written complaint, the product category, the price context and backend checks such as whether the charged price exceeded an allowed range. A practical classifier should use that wider context instead of relying on text alone.

1.3 Problems in Manual Complaint Processing

Manual complaint processing creates several operational problems. It is slow. If administrators must read every complaint one by one, the review queue grows as complaint volume increases. Manual classification can also be inconsistent. Different officials may read similar complaints in different ways, especially when the text is informal or incomplete. Pattern detection becomes difficult too. A digital system can count complaints about expired products or overpricing but those summaries depend on reliable classification.

Another problem is cognitive fatigue. Repeated reading of short complaint texts can lead to shallow review, especially when many entries are similar. A lightweight classifier cannot replace official judgment but it can provide a first-level severity signal. Administrators can then focus human attention on high-priority complaints while still retaining oversight.

1.4 Bengali and Banglish NLP Challenges

Bengali NLP has challenges that do not appear in English-only complaint classification. Bengali has its own script, morphology and punctuation style. Informal digital writing adds spelling variation and inconsistent spacing. In NityMulya, users write complaints in Bengali script, English and Banglish. Banglish means Bengali words written in Latin characters, often mixed with English words. For example, a user may write “dokan beshi taka niyeche” to say that a shop charged too much money.

Banglish is difficult because there is no single standardized spelling. The Bengali word for shop may appear as *dokan*, *dokaan* or mixed with Bengali script. The word for damaged or spoiled may appear as *nosto*, *noshto* or নষ্ট. A word level model may treat these as unrelated tokens. Character n-grams and normalization dictionaries can reduce this problem by capturing subword patterns and mapping frequent Romanized terms to Bengali equivalents.

1.5 Motivation of the Study

The motivation behind this work is practical. Many strong NLP systems now use large transformer models such as BERT and its multilingual or Bengali variants [8,9]. These models can be powerful but they may require more memory, GPU resources, deployment work and maintenance than a small public service platform can support. With only 500 records, a transformer may also overfit unless it is tuned carefully or trained with more data.

For that reason, we study a classical ML pipeline that combines text features with backend structured features. The central question is realistic: if a government marketplace already stores complaint text, product category, validity score and priority score, can those fields be fused into a lightweight decision support classifier? The answer matters for low resource deployment because TF-IDF and linear models are easier to train, explain and run on CPU servers.

1.6 Problem Statement

This research addresses the automatic classification of Bengali, English and Banglish government marketplace complaints into weak, medium and high severity classes. The dataset contains 500 complaints. Text-only classification is difficult because the

complaints are short, noisy and multilingual. Backend numeric features such as validity and priority add structured operational information that can support severity prediction. The task, then, is to design, implement and evaluate a lightweight hybrid classifier that uses both NLP features and backend features without adding new data or reporting invented results.

1.7 Research Objectives

The objectives of this research are:

1. To design a preprocessing pipeline for Bengali, English and Banglish complaint text.
2. To compare word level, character level and combined TF-IDF representations.
3. To integrate text features with backend product category and numeric features.
4. To compare Logistic Regression, Linear SVC, Random Forest and Voting ensemble classifiers.
5. To evaluate class weighting and SMOTE for class imbalance while avoiding data leakage.
6. To perform hyperparameter tuning and stratified five fold cross validation.
7. To conduct an ablation study showing the contribution of text, product category and numeric backend features.
8. To present a thesis level analysis of results, limitations and deployment implications.

1.8 Research Questions

The study asks the following research questions:

1. How effective are classical TF-IDF based models for Bengali, English and Banglish complaint severity classification?
2. Do character n-grams improve robustness against Banglish spelling variation and mixed script text?

3. How much do backend validity and priority features contribute compared with text-only features?
4. Which classifier performs best on sparse TF-IDF and structured hybrid features?
5. Does SMOTE improve minority class performance when applied only inside training folds?
6. What are the deployment advantages and risks of using backend derived features in a government marketplace classifier?

1.9 Scope of the Study

The scope is limited to severity classification of complaints collected from the NityMulya marketplace platform. The dataset contains 500 records and three target classes: weak, medium and high. We do not attempt full legal complaint adjudication, automatic punishment recommendation, named-entity extraction or multilingual neural translation. The focus is lightweight classification using classical machine learning.

The model also does not claim to understand complaint meaning in the same way a human officer does. The strongest model uses backend numeric features and these features are closely related to the operational rules used in severity assessment. The system should be read as a decision support component that combines language signals with backend rule derived signals.

1.10 Research Contribution

The main contributions are:

1. A reproducible low resource complaint classification pipeline for Bengali, English and Banglish government marketplace complaints.
2. A hybrid feature fusion design that combines normalized text, product category, validity score and priority score.
3. A systematic comparison of word TF-IDF, character TF-IDF, combined TF-IDF, Logistic Regression, Linear SVC, Random Forest, Voting ensemble, class weighting and SMOTE.

4. A clear ablation study demonstrating that backend numeric features are the major contributors to performance while text alone is insufficient for high accuracy in this dataset.
5. A deployment oriented discussion explaining why lightweight classical ML can be suitable for CPU friendly public service systems.

1.11 Thesis Organization

Chapter 2 reviews literature on NLP, text classification, Bengali and Banglish language processing, TF-IDF, classical classifiers, imbalance handling and e-governance complaint systems. Chapter 3 describes the NityMulya platform, backend architecture, database design, dataset properties, label definitions and ethical considerations. Chapter 4 explains the methodology, including preprocessing, feature engineering, model training, cross validation, tuning, SMOTE and ablation. Chapter 5 presents implementation details. Chapter 6 reports and analyzes the results. Chapter 7 discusses practical implications and reviewer concerns. Chapter 8 describes limitations and future work. Chapter 9 concludes the thesis. The appendices provide sample data, normalization dictionary entries, search spaces, reports, code snippets, schema examples and additional visualizations.

Chapter 2

Literature Review

2.1 Introduction to Natural Language Processing

Natural Language Processing is a field of artificial intelligence that studies how computers process human language. In classification tasks, NLP methods convert raw text into numerical representations that a machine learning model can use. Traditional NLP often relies on tokenization, stopword removal, n-grams, frequency statistics and classical classifiers. Modern NLP uses neural embeddings and transformer models more often but classical methods are still useful when data and computing resources are limited.

For complaint classification, the main NLP task is to map an unstructured text field to a structured class label. A sentence such as “The shop charged more than the listed price” carries a clear overpricing signal. Its severity, however, may also depend on price rules and backend validation. For government marketplaces, a classifier should be judged as part of a larger decision support pipeline, not only as a language model.

2.2 Text Classification Systems

Text classification assigns one or more labels to a document. Common uses include spam detection, sentiment analysis, topic classification, customer support routing, news categorization and grievance triage. A typical pipeline has four stages: text cleaning, feature extraction, model training and evaluation. Weakness in any one stage can reduce the final result.

In classical text classification, the feature matrix is usually sparse and high-

dimensional. Each column may represent a word, phrase or character sequence while each row represents a document. Linear models are strong baselines for this kind of data because they can learn class boundaries directly in sparse vector spaces [2]. Tree-based models often work well on dense tabular data but sparse text matrices are less friendly to axis-aligned splitting.

2.3 Bengali NLP Research

Bengali is one of the most widely spoken languages in the world, yet NLP resources for Bengali are still less mature than English resources. Bengali text processing needs script aware normalization, tokenization and careful vocabulary handling. Researchers have worked on Bengali sentiment analysis, news classification, hate speech detection and language model pretraining. BanglaBERT and related transformer models show that pretraining on Bengali corpora can improve language understanding tasks [9].

Many Bengali NLP studies use larger public corpora, social media datasets or news datasets. Government marketplace complaints have a different character. They are short, domain specific and connected to backend records such as product categories and price validations. This leaves a gap for practical Bengali complaint classification in administrative systems.

2.4 Banglish Language Processing Challenges

Banglish is a code mixed and transliterated form of Bengali written with Latin characters, often mixed with English and Bengali script. It has no fixed spelling convention. A user may write “beshi taka”, “beshi tk” or “bashi product” in the same complaint space. Such variation weakens word level matching and increases vocabulary sparsity.

Banglish can be processed through transliteration, dictionary based normalization, subword features or multilingual embeddings. In a small data setting, a simple dictionary is easier to inspect than a neural transliteration model. Character n-grams also help because they preserve partial spelling patterns. For example, *dokan*, *dokan-dar* and *dokaan* share character sequences that may help a classifier generalize.

2.5 TF-IDF Feature Extraction

Term Frequency-Inverse Document Frequency is a statistical representation that assigns higher weights to terms that are frequent in a document but less common across the corpus [1]. The term frequency component captures local importance while the inverse document frequency component reduces the influence of very common terms.

For a term t in document d , term frequency may be expressed as:

$$TF(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}, \quad (2.1)$$

where $f_{t,d}$ is the frequency of term t in document d . The inverse document frequency can be written as:

$$IDF(t) = \log \left(\frac{N}{1 + df(t)} \right), \quad (2.2)$$

where N is the number of documents and $df(t)$ is the number of documents containing the term. The TF-IDF weight is:

$$TFIDF(t, d) = TF(t, d) \times IDF(t). \quad (2.3)$$

TF-IDF suits this study because it is transparent, fast and effective with small datasets. It does not require GPU training or large external corpora.

2.6 Character N-Grams

An n -gram is a sequence of n items. In word n -grams, the items are words. In character n -grams, the items are characters. Character n -grams are useful in noisy multilingual text because they can represent spelling fragments and morphological patterns. The `char_wb` analyzer used in this study creates character n -grams inside word boundaries, reducing noise from cross-word character sequences.

For Bengali and Banglish complaints, character n -grams help with transliteration variation, typographical errors and mixed scripts. In our experiment, character `char_wb` n -grams with range $(3, 5)$ achieved the strongest TF-IDF variant result under Linear SVC.

2.7 Logistic Regression for NLP

Logistic Regression is a linear classifier that estimates class probabilities from weighted feature sums. For binary classification, the probability of class 1 is:

$$P(y = 1|x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}}. \quad (2.4)$$

For multiclass classification, Logistic Regression extends this idea through one-vs-rest or multinomial formulations. In text classification, Logistic Regression is effective because sparse TF-IDF features often become linearly separable enough for useful decision boundaries. It also provides regularization through the parameter C , where smaller C means stronger regularization.

2.8 Support Vector Machines for Text Classification

Support Vector Machines learn a separating hyperplane that maximizes the margin between classes [3]. Linear SVC is widely used for high-dimensional text features because it focuses on boundary examples and scales well with sparse matrices. The hinge loss objective can be written as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max(0, 1 - y_i(w^T x_i + b)). \quad (2.5)$$

The regularization parameter C controls the trade-off between maximizing margin and reducing classification errors.

2.9 Random Forest for Classification

Random Forest is an ensemble of decision trees trained with bootstrap sampling and random feature selection [4]. It often works well for tabular data with nonlinear interactions. Sparse and high-dimensional TF-IDF matrices are harder for tree splits. A Random Forest may need many trees to model useful boundaries and it can still fall behind linear models on sparse text. Our results followed this pattern: Random Forest achieved much lower weighted F1 than Logistic Regression and Linear SVC.

2.10 SMOTE and Imbalance Handling

Class imbalance occurs when one class has many more examples than another. In this dataset, the medium class is the majority class while weak has fewer records. If imbalance is ignored, a classifier may overpredict the majority class. Class weighting adjusts the loss so minority classes receive higher penalty. SMOTE creates synthetic minority examples by interpolating between minority samples [5]. The synthetic sample can be represented as:

$$x_{new} = x_i + \lambda(x_{nn} - x_i), \quad (2.6)$$

where x_i is a minority sample, x_{nn} is one of its nearest minority neighbors and $\lambda \in [0, 1]$.

SMOTE must be applied only to training data. If synthetic samples are generated before cross validation splitting, information from validation folds can leak into training folds. We avoided that problem by placing SMOTE inside the `imblearn` pipeline.

2.11 E-Governance Complaint Systems

E-governance complaint systems aim to improve administrative response, transparency and accountability. A complaint platform can help administrators identify urgent issues, monitor repeated violations and allocate resources. Trust is essential in this setting. Automated classification should support human decision making rather than silently replace administrative review.

In marketplace regulation, backend data is especially relevant. Product category, expected price range, complaint history, validity score and priority score can all provide structured evidence. A hybrid approach is more realistic than a text-only model.

2.12 Existing Bengali Complaint Classification Works

Compared with English customer service classification, Bengali government complaint classification is less represented in published literature. Existing Bengali NLP studies

provide useful methods for representation learning and classification but most are not designed specifically for government marketplace triage. Table 2.1 summarizes representative work and positions the present thesis.

Table 2.1: Comparison with representative related work

Study	Focus	Method	Relevance and gap
Salton and Buckley [1]	Term weighting	TF-IDF	Establishes the statistical representation used in this thesis.
Joachims [2]	Text categorization	SVM on sparse text	Supports the use of linear models for high-dimensional text data.
Chawla et al. [5]	Imbalanced learning	SMOTE	Provides the oversampling method evaluated in this study.
Devlin et al. [8]	Language modeling	BERT transformer	Shows strength of transformers but deployment cost is higher.
Bhattacharjee et al. [9]	Bengali language understanding	BanglaBERT	Important Bengali transformer baseline but not the selected lightweight approach.
This thesis	Bengali/Banglish marketplace complaints	TF-IDF + backend features + classical ML	Focuses on low resource, deployable hybrid decision support.

2.13 Research Gaps

The literature review points to four gaps. First, Bengali and Banglish complaint classification for government marketplaces remains underexplored. Second, many NLP studies focus on text alone while real complaint platforms also store structured backend evidence. Third, transformer models are powerful but may be unnecessary or costly for small datasets and public service deployment. Fourth, many studies report performance without detailed ablation analysis, which makes the role of each feature

group unclear.

This work addresses these gaps through a transparent hybrid pipeline, cross validation, hyperparameter tuning, imbalance experiments and an ablation study.

Chapter 3

System Overview and Dataset

3.1 NityMulya Platform Overview

NityMulya is a government marketplace oriented platform for citizens, shops and administrators involved in essential commodity monitoring. The complaint module lets citizens report issues related to price, product quality, expiry, weight, packaging and shop behavior. The platform stores complaint records in a Supabase PostgreSQL backend, so the data can be used for analysis and administrative review.

The system studied here is not only a text classifier. It belongs to a broader marketplace workflow where complaint text is connected to customer identity, shop identity, product subcategory and backend rule-based scores. This context matters because it explains why backend features improve classification performance.

3.2 System Workflow

The workflow begins when a citizen submits a complaint through the application. The backend stores the complaint, links it to product and shop information, computes validity and priority scores and assigns an operational severity class. The machine learning pipeline then exports the required fields, preprocesses them, trains models and evaluates classification performance.

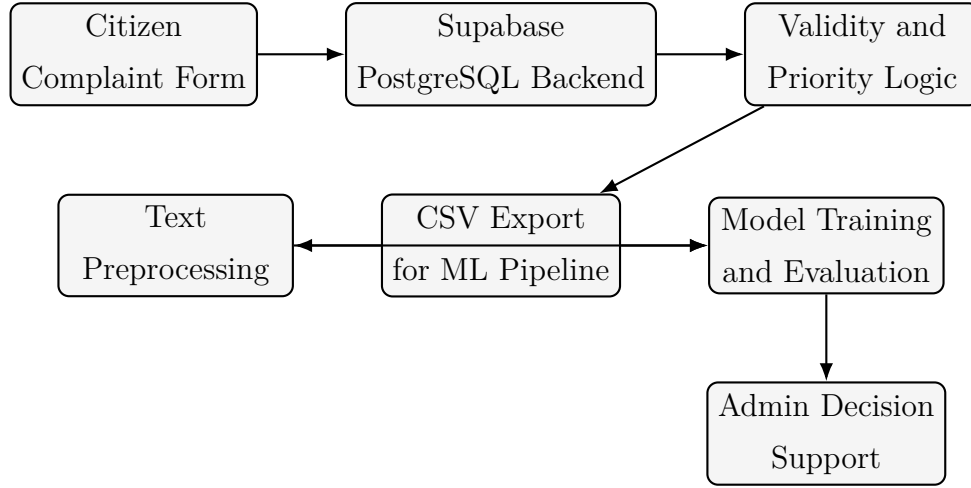


Figure 3.1: High-level workflow of the NityMulya complaint classification system

3.3 Supabase Backend Architecture

Supabase provides a hosted PostgreSQL database with API access, authentication and storage features. For the complaint module, PostgreSQL acts as the system of record. Complaint entries are stored with identifiers, timestamps, customer information, shop information, product subcategory, complaint text, validity score, priority score, complaint score and final classification.

The backend architecture matters because the ML model uses features that are available at operational time. Validity and priority are not generated by the ML model. They are backend scores produced by marketplace rules and keyword logic. In deployment, a new complaint would first pass through the backend scoring logic and then through the lightweight classifier.

3.4 PostgreSQL Storage Design

The complaint table stores both unstructured and structured data. The complaint description is the unstructured part. Product subcategory, validity, priority and complaint score are structured fields. Figure 3.2 shows the simplified schema.

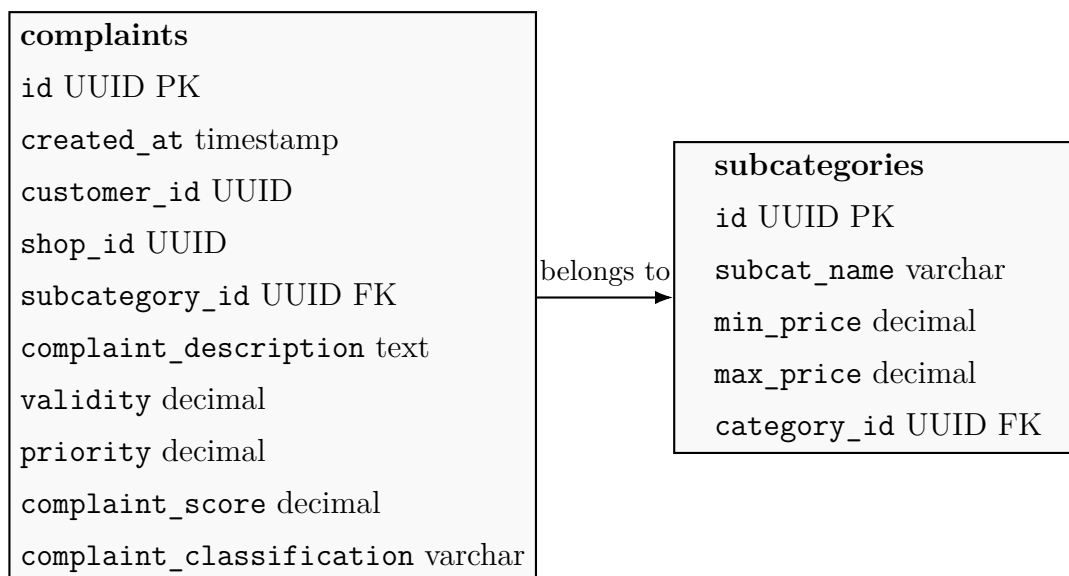


Figure 3.2: Simplified database schema used by the complaint dataset

3.5 Complaint Submission Process

The complaint submission process has several steps. A citizen selects or enters relevant marketplace information, chooses the product category or subcategory, writes a complaint description and submits the form. The backend stores the entry and computes rule-based values. Price-related complaints may be compared with product price ranges. Quality or safety complaints may receive higher priority when they contain terms related to expiry, health hazard, fraud or counterfeit goods.

3.6 Dataset Collection Pipeline

We exported the dataset from the backend into a CSV file named `complaints_full.csv`. The machine learning pipeline loads the CSV, drops records with missing required fields, maps target labels to numeric classes and separates features from labels. The required fields are complaint description, complaint classification, subcategory name, validity and priority.

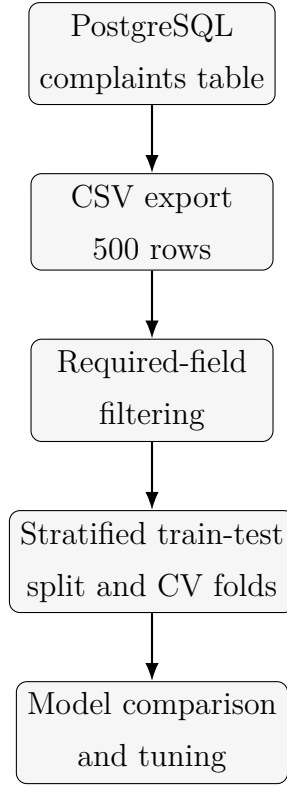


Figure 3.3: Dataset collection and experimental preparation flow

3.7 Dataset Characteristics

The dataset contains 500 complaint entries. The class distribution is imbalanced: 70 weak complaints, 310 medium complaints and 120 high complaints. The medium class accounts for 62.0% of the dataset. The average validity score is 0.6223, the average priority score is 0.5235 and the average complaint score is 0.5826. Validity ranges from 0.10 to 0.95, priority ranges from 0.20 to 0.95 and complaint score ranges from 0.16 to 0.91.

Table 3.1: Dataset class distribution

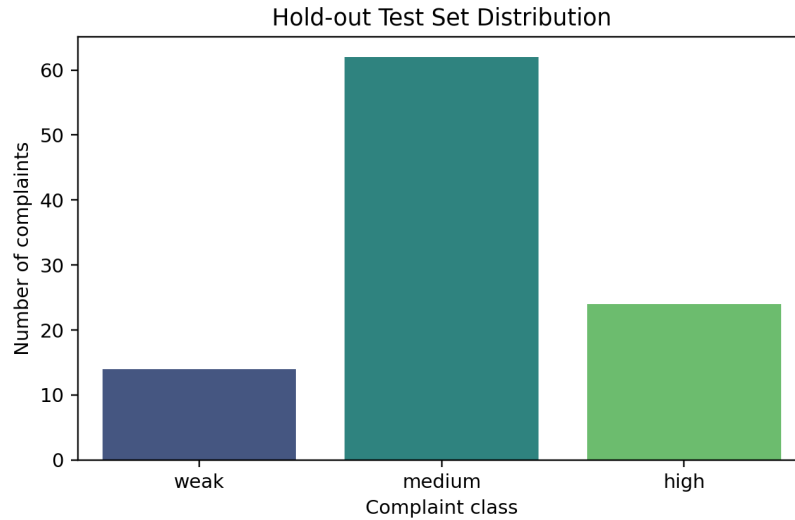
Class	Count	Percentage
Weak	70	14.0%
Medium	310	62.0%
High	120	24.0%

Table 3.2: Backend numeric feature summary

Feature	Minimum	Maximum	Mean
Validity	0.10	0.95	0.6223
Priority	0.20	0.95	0.5235
Complaint score	0.16	0.91	0.5826

3.8 Data Distribution

The stratified hold-out test set contains 100 complaints: 14 weak, 62 medium and 24 high. This mirrors the overall class distribution and prevents the test split from accidentally overrepresenting or underrepresenting any severity class.

**Figure 3.4:** Distribution of complaint classes in the 20% stratified hold-out test set

3.9 Label Definition

The target label has three classes. Weak complaints represent low-severity or low-validity cases. Medium complaints represent cases that require attention but may not demand immediate action. High complaints represent cases with stronger validity, urgency or public-risk signals.

The backend documentation defines complaint score as:

$$ComplaintScore = 0.6 \times Validity + 0.4 \times Priority. \quad (3.1)$$

Operational labels are assigned using score thresholds: high for scores greater than or equal to 0.70, medium for scores from 0.40 to 0.69 and weak for scores below 0.40. This relationship matters for interpretation. Since validity and priority help define the operational label, they naturally carry strong predictive information. They are appropriate for a backend decision support model but the result should not be read as proof that text alone solves the task.

3.10 Complaint Examples

Table 3.3 presents representative examples showing Bengali, English and Banglish variation. Personal identifiers are omitted.

Table 3.3: Representative complaint examples

Language	Complaint text	Class
Bengali	সয়াবিন তেল এর গুণমান খারাপ ছিল। পচা এবং দুর্গন্ধযুক্ত।	High
English	The product quality was very poor. Rotten and smelly.	High
Banglish	Dokane expired dim bikri kore. Swasther jonno khotikar.	High
Mixed	Bashi morich bikri koreche. Taka ferot chay nai.	Medium

3.11 Ethical Considerations

Complaint data can contain personal information, shop names and allegations. A complaint classification system must be designed with privacy and fairness in mind. This study uses exported research data for model evaluation and does not require disclosure of personally identifiable information in the thesis text. Examples are limited and should be anonymized in any public version.

The classifier should not be used as the sole authority for legal or punitive decisions. It should support prioritization and routing while human administrators retain responsibility for final decisions. This is particularly important when model predictions are wrong or when a complaint contains unusual details not captured by the training data.

3.12 Data Privacy Discussion

Data privacy requires three practical steps: minimize unnecessary personal data, restrict access to raw complaints and log model outputs responsibly. If deployed, the system should store predictions with timestamps and model version identifiers so that decisions can be audited. Sensitive fields such as customer identifiers and contact details should not be used in the ML feature set unless there is a clear legal and ethical basis. This study uses complaint text, product subcategory, validity and priority only.

Chapter 4

Methodology

4.1 Overall Machine Learning Pipeline

The methodology follows a structured pipeline: data loading, preprocessing, feature extraction, feature fusion, model training, cross validation, hyperparameter tuning, final hold-out evaluation and result reporting. Figure 4.1 gives the full flow.

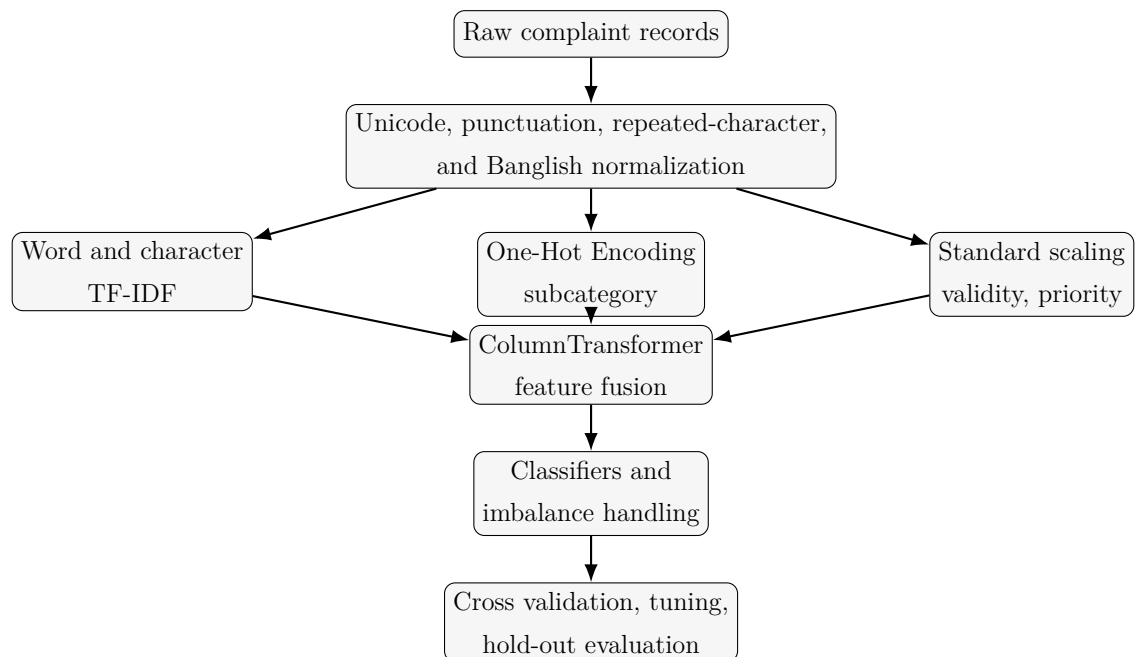


Figure 4.1: Machine learning pipeline used in the thesis

4.2 Text Preprocessing Pipeline

We designed the preprocessing pipeline for Bengali, English and Banglish complaint text. The steps are:

1. Convert missing values to empty strings.
2. Apply Unicode NFKC normalization.
3. Lowercase English and Banglish tokens.
4. Compress repeated characters.
5. Remove noisy punctuation while preserving Bengali, Latin letters, digits and spaces.
6. Replace frequent Banglish marketplace terms using a dictionary.
7. Optionally remove Bengali stopwords.

The goal is not to over-clean the text. Too much cleaning can remove useful complaint signals. The pipeline removes common noise while preserving domain words such as product names, price terms, expiry terms and quality terms.

4.3 Unicode Normalization

Unicode normalization converts visually similar character sequences into a consistent representation. We use NFKC normalization. In multilingual text, different input methods can create compatibility characters or inconsistent forms. Normalization reduces unnecessary variation before tokenization and vectorization.

4.4 Banglish Normalization

Banglish normalization maps frequent Romanized marketplace terms to Bengali equivalents. Examples include *dokan* to দোকান, *dam* to দাম, *taka* to টাকা, *chal* to চাল and *nosto* to নষ্ট. The dictionary is simple by design, so it can be checked and revised easily. A larger neural transliteration system may be useful later but this work prioritizes transparency and low resource deployment.

4.5 Stopword Removal

Stopwords are frequent function words that may carry less class-specific information. We implemented Bengali stopwords removal as an optional parameter and the final Logistic Regression model selected it during hyperparameter tuning. Stopword removal is not always helpful in short texts because even common words can support phrase patterns. For that reason, we treated it as a tunable choice rather than a fixed rule.

4.6 TF-IDF Mathematics

TF-IDF transforms text into numeric vectors by combining local term frequency with corpus-level rarity. We evaluated both word level and character level TF-IDF features. Sublinear term frequency is used so that repeated words do not dominate linearly. If tf is the raw term frequency, sublinear scaling uses:

$$tf_{sublinear} = 1 + \log(tf). \quad (4.1)$$

The final TF-IDF matrix is sparse because each complaint contains only a small subset of possible words or character n-grams. Sparse matrices are memory-efficient and work well with linear classifiers.

4.7 Word N-Gram Extraction

Word n-grams represent sequences of words. A unigram model uses individual words while a bigram model includes adjacent two-word phrases. For example, “expired product” and “beshi taka” can be more informative than the words alone. We compared word n-gram ranges $(1, 1)$ and $(1, 2)$.

4.8 Character N-Gram Extraction

Character n-grams represent subword sequences. The character `char_wb` n-gram range $(3, 5)$ is used because three-to-five-character fragments can capture meaningful spelling patterns without producing excessively long sequences. Character n-grams are particularly helpful for Banglish spelling variation and incomplete words.

4.9 FeatureUnion Explanation

When word and character TF-IDF are combined, the pipeline uses a **FeatureUnion**. A **FeatureUnion** runs multiple feature extractors in parallel and concatenates their outputs. If X_w is the word TF-IDF matrix and X_c is the character TF-IDF matrix, the combined text matrix is:

$$X_{text} = [X_w \ X_c]. \quad (4.2)$$

This allows the model to use both complete-token meaning and subword robustness.

4.10 Numeric Feature Scaling

The backend numeric features are validity and priority. These values are scaled using **StandardScaler** with sparse compatibility. Scaling is useful because models such as Logistic Regression and Linear SVC are sensitive to feature magnitudes. If μ is the mean and σ is the standard deviation, scaling is:

$$z = \frac{x - \mu}{\sigma}. \quad (4.3)$$

4.11 One-Hot Encoding

Product category or subcategory is categorical. One-Hot Encoding converts each category into a binary feature. Unknown categories are ignored during transformation, so the model can process categories that were not seen during training. In our ablation study, product category did not improve this dataset and reduced weighted F1 when added to text.

4.12 Cross Validation Methodology

Stratified five fold cross validation was used to estimate model performance. Stratification preserves class proportions in each fold. For each fold, the model trains on four parts and validates on the remaining part. If the fold metric values are $m_1, m_2, \dots, m_5$, the mean and standard deviation are:

$$\bar{m} = \frac{1}{5} \sum_{i=1}^5 m_i, \quad (4.4)$$

$$s = \sqrt{\frac{1}{5} \sum_{i=1}^5 (m_i - \bar{m})^2}. \quad (4.5)$$

4.13 Hyperparameter Tuning

We used `RandomizedSearchCV` to tune candidate models. Randomized search samples combinations from a defined parameter space and evaluates them with cross validation. It is more efficient than exhaustive grid search when several hyperparameters exist. The tuned parameters included TF-IDF maximum features, word n-gram range, optional Bengali stopwords removal, Logistic Regression C , Linear SVC C and Random Forest tree parameters.

4.14 Class Imbalance Handling

The dataset is imbalanced and the medium class is the majority class. Logistic Regression and Linear SVC used `class_weight='balanced'`, which increases the loss contribution of minority classes. We also tested SMOTE. To avoid leakage, we placed SMOTE inside the training pipeline so that synthetic samples were created only from the training portion of each fold.

4.15 Ablation Study Methodology

An ablation study removes or adds feature groups to measure their contribution. Four feature sets were evaluated:

1. Text only.
2. Text plus product category.
3. Text plus backend numeric features.
4. Full hybrid feature set: text, product category and numeric features.

The ablation study is central to this work because it shows that backend numeric features are the main reason performance improves.

4.16 Why Linear Models Outperform Random Forest on Sparse TF-IDF

Sparse TF-IDF vectors have many dimensions and many zero values. Linear models can assign weights directly to useful terms and learn a separating hyperplane. Random Forest models split features one at a time, so they may struggle when useful evidence is spread across many sparse dimensions. In small datasets, tree ensembles may also overfit local patterns or fail to find stable splits. This helps explain why Linear SVC and Logistic Regression outperformed Random Forest in the experiments.

4.17 Pseudocode

Listing 4.1: Pseudocode for the experimental pipeline

```
Input: Complaint dataset D with text, subcategory, validity, priority,
      label
Output: Evaluation tables, final classifier, confusion matrix

1. Load D from complaints_full.csv
2. Drop rows with missing required fields
3. Map labels: weak -> 0, medium -> 1, high -> 2
4. Create stratified 80/20 train-test split
5. For each TF-IDF variant:
    preprocess text
    extract features
    evaluate Linear SVC with 5-fold StratifiedKfold
6. For each classifier and imbalance setting:
    build hybrid feature pipeline
    optionally apply SMOTE inside training folds
    evaluate with 5-fold StratifiedKfold
7. Tune Logistic Regression, Linear SVC and Random Forest
8. Select best tuned model family by weighted F1
9. Run ablation study using best tuned settings
10. Fit final model on training split
11. Evaluate final model on hold-out test split
12. Save reports, tables and visualizations
```

Chapter 5

Implementation

5.1 Development Environment

We developed the implementation in Python using a CPU friendly machine learning stack. The main training script is located at `ML_Model_Training/src/pipeline.py`. The dataset is stored at `ML_Model_Training/data/complaints_full.csv` and the generated metrics and figures are stored in `ML_Model_Training/results/`.

5.2 Python Ecosystem

We selected Python because it provides strong libraries for data processing, machine learning, evaluation and visualization. The main libraries are Scikit-learn, Imbalanced-learn, NumPy, Matplotlib, Seaborn and Pandas. Scikit-learn provides the pipeline, vectorizer, classifier and cross validation tools. Imbalanced-learn provides SMOTE and pipeline support for leakage-safe oversampling.

5.3 Scikit-Learn

We used Scikit-learn for `TfidfVectorizer`, `ColumnTransformer`, `FeatureUnion`, `LogisticRegression`, `LinearSVC`, `RandomForestClassifier`, `VotingClassifier`, `StratifiedKFold`, `cross_validate` and `RandomizedSearchCV`. Pipelines are important here because preprocessing, feature extraction, scaling, oversampling and classification are handled as one reproducible object.

5.4 Imbalanced-Learn

Imbalanced-learn provides the pipeline that can place SMOTE between feature extraction and classification. This is needed because Scikit-learn's standard pipeline does not treat oversampling as a typical transformer. By placing SMOTE inside the cross validation workflow, the implementation prevents validation data from influencing synthetic training samples.

5.5 Supabase Integration

Supabase PostgreSQL stores the complaint data. We used the exported CSV for offline model training and evaluation. In a deployment setting, a backend service can load the trained model and classify new complaints after the backend computes validity and priority. The Supabase table remains the source of operational data while the ML model adds decision support predictions.

5.6 Data Loading

The data-loading function reads the CSV file, checks required columns, removes incomplete rows, maps class labels to integers and returns the feature and target matrices. The relevant feature columns are complaint description, subcategory name, validity and priority.

Listing 5.1: Data-loading structure from the training pipeline

```
required_columns = [
    "complaint_description",
    "complaint_classification",
    "subcategory_name",
    "validity",
    "priority",
]
df = df.dropna(subset=required_columns).copy()
df["target"] = df["complaint_classification"].map(CLASS_MAPPING)
X = df[["complaint_description", "subcategory_name", "validity", "priority"]
      ]
y = df["target"]
```

5.7 Pipeline Implementation

The feature transformer uses `ColumnTransformer`. The text column is processed by the normalizer and TF-IDF pipeline. The product subcategory is processed by One-Hot Encoding when included. Validity and priority are scaled when numeric features are included.

Listing 5.2: Simplified hybrid feature transformer

```
transformers = [  
    ("text", build_text_pipeline(...), "complaint_description")  
]  
  
if feature_set in {"text_category", "full"}:  
    transformers.append(  
        ("category", OneHotEncoder(handle_unknown="ignore"), ["  
            subcategory_name"])  
    )  
  
if feature_set in {"text_numeric", "full"}:  
    transformers.append(  
        ("numeric", StandardScaler(with_mean=False), ["validity", "priority  
            "])  
    )
```

5.8 Model Training Implementation

Each model is wrapped in the same pipeline interface. This keeps preprocessing and feature representation comparable across classifiers. Logistic Regression and Linear SVC use balanced class weights. Random Forest is tested with class weighting and tree hyperparameters.

5.9 Evaluation Implementation

The evaluation reports accuracy, weighted precision, weighted recall and weighted F1. Weighted metrics fit this dataset because the classes are imbalanced and each class should contribute according to its support. The final hold-out report also includes

class-wise precision, recall and F1.

5.10 Visualization Generation

The pipeline generates a class distribution plot, confusion matrix, two-dimensional SVD projection of transformed features and K-Means cluster projection. These figures are used in Chapter 6 to support interpretation. The SVD plots are exploratory visualizations, not primary evidence of classifier performance.

5.11 Training Workflow Diagram

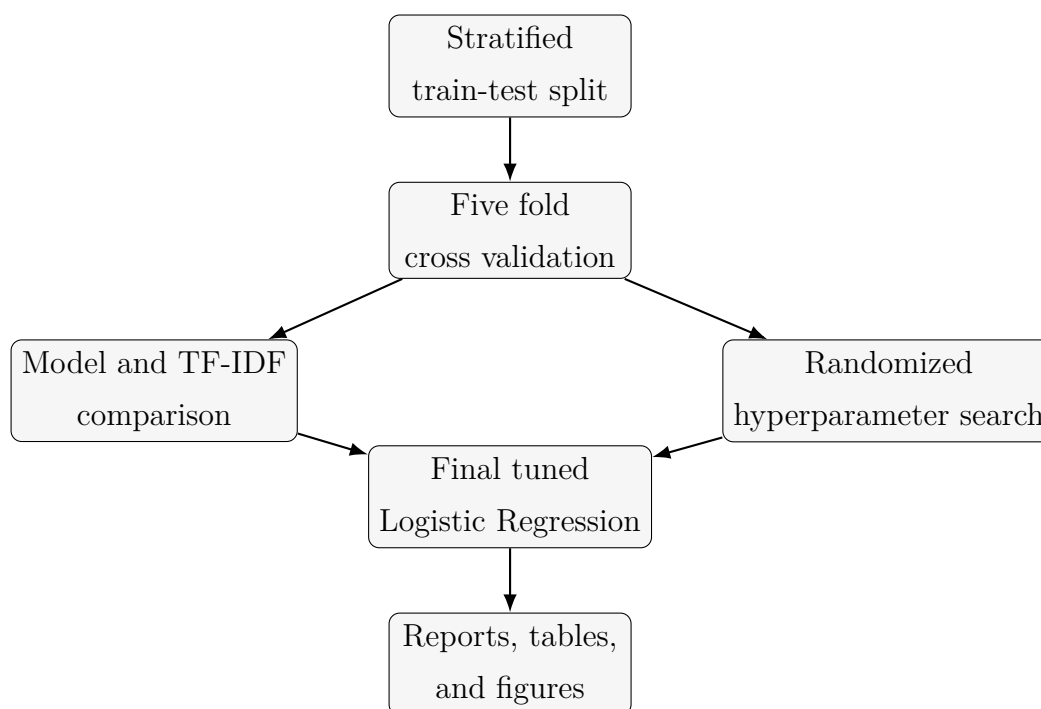


Figure 5.1: Training and evaluation workflow

5.12 Deployment Possibilities

The trained model can be deployed as a backend service in several ways. It can run as a Python microservice called from the Node.js or application backend. It can also be serialized with `joblib` and loaded into a scheduled batch job that classifies new complaints. Because the selected model is Logistic Regression with TF-IDF and two numeric features, inference is lightweight and CPU friendly. This makes it suitable for modest cloud instances and government systems where GPU access is limited.

Chapter 6

Results and Analysis

6.1 Evaluation Metrics

We use accuracy, weighted precision, weighted recall and weighted F1. Precision measures how many predicted examples of a class are correct. Recall measures how many actual examples of a class are found. F1 is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \quad (6.1)$$

Weighted F1 averages class F1 scores by class support, making it suitable for imbalanced datasets.

6.2 Baseline Random Forest Results

The original Random Forest baseline achieved 58.00% accuracy and 0.58 weighted F1 on the same 20% hold-out test split. The weak class F1 was approximately 0.30. This result showed that the initial model was not sufficient for reliable triage, especially for minority class behavior.

After improving preprocessing, feature extraction, model comparison, imbalance handling, tuning and ablation, the best final model achieved much stronger performance. This gain came from model and feature design, not from adding more data.

6.3 Improved Logistic Regression Results

The final selected model was tuned Logistic Regression using normalized text and backend numeric features. It achieved 86.00% hold-out accuracy and 0.8594 weighted F1. Table 6.1 gives the class-wise performance.

Table 6.1: Final hold-out classification report for tuned Logistic Regression

Class	Precision	Recall	F1-score	Support
Weak	0.9231	0.8571	0.8889	14
Medium	0.8750	0.9032	0.8889	62
High	0.7826	0.7500	0.7660	24
Macro average	0.8602	0.8368	0.8479	100
Weighted average	0.8596	0.8600	0.8594	100

The weak and medium classes achieved strong F1 scores while the high class remained more difficult. The high class had 24 examples in the test split and six high complaints were misclassified as medium. This suggests that the boundary between medium and high remains a practical challenge.

6.4 TF-IDF Comparison Analysis

Table 6.2 presents the five fold TF-IDF comparison using Linear SVC.

Table 6.2: Five fold TF-IDF comparison using Linear SVC

Feature extractor	Accuracy	Weighted F1	F1 std.
Character <code>char_wb</code> (3,5)	0.8220	0.8225	0.0286
Word unigram + bigram	0.8220	0.8209	0.0359
Word + character	0.8100	0.8092	0.0342
Word unigram	0.8040	0.8041	0.0432

Character n-grams achieved the highest weighted F1 among TF-IDF variants. This supports the idea that subword features help with spelling variation, Banglish

text and mixed script complaints. Word unigrams and bigrams also performed competitively, which suggests that phrase-level signals such as “expired product” or “beshi taka” are useful.

6.5 Model Comparison Analysis

Table 6.3 reports the classifier and imbalance comparison on the full hybrid feature set.

Table 6.3: Five fold model comparison on the full hybrid feature set

Classifier	Imbalance handling	Accuracy	Weighted F1	F1 std.
Linear SVC	Class weight	0.8100	0.8092	0.0342
Voting Ensemble	Class weight	0.8020	0.8013	0.0314
Logistic Regression	Class weight	0.7980	0.8000	0.0291
Linear SVC	Class weight + SMOTE	0.7820	0.7814	0.0378
Voting Ensemble	Class weight + SMOTE	0.7580	0.7601	0.0320
Logistic Regression	Class weight + SMOTE	0.7400	0.7461	0.0344
Random Forest	Class weight	0.6460	0.6224	0.0492
Random Forest	Class weight + SMOTE	0.6160	0.6083	0.0309

Linear SVC achieved the best default cross validation score. After tuning, Logistic Regression was selected because it produced the strongest tuned family and the best final hold-out behavior with the selected feature set. Random Forest performed poorly compared with linear models, which matches the sparse TF-IDF explanation in Chapter 4.

6.6 Cross Validation Analysis

Cross validation gives a more reliable estimate than a single split because every record is used for validation once. The best default row, Linear SVC without SMOTE, achieved 0.8092 weighted F1 with a standard deviation of 0.0342. The tuned Logistic Regression family achieved 0.8093 weighted F1 during tuning and 0.8249 full-hybrid cross validation weighted F1 when evaluated using the tuned configuration.

The difference between cross validation and hold-out performance needs careful reading. The hold-out test set has only 100 examples, so the exact value can vary depending on the split. The stronger conclusion is the repeated pattern: linear models

perform better than Random Forest and numeric backend features greatly improve performance.

6.7 SMOTE Analysis

SMOTE did not improve the results. In every classifier family, SMOTE reduced weighted F1 compared with class weighting alone. A likely reason is that synthetic interpolation in sparse high-dimensional feature space creates artificial vectors that do not match realistic complaint patterns. For small text datasets, class weighting is often safer than oversampling. We still tested SMOTE correctly by applying it only within training folds.

6.8 Ablation Study Analysis

The ablation study is the key result for understanding the system. Table 6.4 shows that text alone achieved only 0.5559 weighted F1. Adding product category reduced performance slightly to 0.5410. Adding numeric backend features increased weighted F1 to 0.8503. The full feature set achieved 0.8249, which is lower than text plus numeric features.

Table 6.4: Ablation study with tuned Logistic Regression

Feature set	Accuracy	Weighted F1	F1 std.
Text + numeric features	0.8500	0.8503	0.0299
Full hybrid feature set	0.8260	0.8249	0.0247
Text only	0.5580	0.5559	0.0419
Text + product category	0.5400	0.5410	0.0328

This result shows that complaint text alone did not achieve high performance. The text contains useful signals but many complaints are short, repetitive and semantically overlapping. For example, both medium and high complaints may mention poor quality or expired products. Backend validity and priority scores provide structured information that separates severity levels more effectively.

6.9 Why Backend Numeric Features Improved Results

Validity and priority improved results because they represent domain knowledge already computed by the backend. Validity captures whether the complaint appears legitimate according to price or complaint rules. Priority captures urgency, health risk, fraud terms or quality concern. These scores reduce ambiguity in text. A short complaint saying “bad product” may be difficult to classify from text alone but if the backend priority is high and validity is high, the model can assign a more appropriate severity.

This finding must still be read carefully. Validity and priority contribute to the operational complaint score used to define labels, so they are not independent evidence in the same way raw text is. They are legitimate deployment-time features but they also mean the model partly learns the backend severity policy. That is acceptable for a decision support system whose goal is consistent operational triage but it limits any claim about pure NLP understanding.

6.10 Why Product Category Features Reduced Performance

Product category features reduced performance in the ablation study. Several reasons are possible. First, the dataset has many product subcategories compared with only 500 records, so category features may be sparse and unstable. Second, severity may depend less on the product type and more on validity and urgency. Third, some product categories appear in all severity classes, making category a weak predictor. Finally, category features may introduce noise that competes with stronger numeric features.

6.11 Confusion Matrix Analysis

Figure 6.1 shows the final confusion matrix. The model correctly classified 12 of 14 weak complaints, 56 of 62 medium complaints and 18 of 24 high complaints. Most errors occurred between medium and high classes.

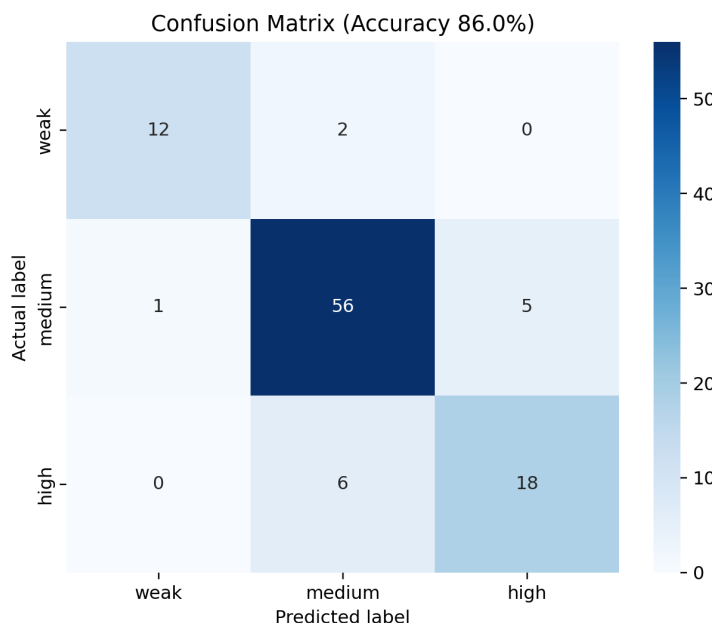


Figure 6.1: Confusion matrix for the final tuned Logistic Regression model

The confusion matrix suggests that the model rarely confuses weak complaints with high complaints. That is useful for triage because the most serious error would be assigning a high-severity complaint to weak. In the hold-out result, no high complaint was predicted as weak. Six high complaints were predicted as medium, so medium-high separation still needs improvement.

6.12 Feature Importance Discussion

For linear models, feature weights can provide interpretability. Positive weights for a class indicate that a feature increases the model's confidence for that class. In this system, backend numeric features are expected to have strong weights because they correlate with severity. Text features related to expiry, health risk, fake products, poor quality, overpricing and weight manipulation may also contribute.

Unlike Random Forest feature importance, linear model coefficients are more natural for sparse TF-IDF. Future work could export class-wise top weighted terms to help administrators understand model decisions.

6.13 Error Analysis

Errors mainly occur where complaint severity is ambiguous. A complaint may use strong language but have low backend validity. Another complaint may have valid pricing evidence but mild wording. Some Banglish complaints may contain uncommon spellings not covered by the dictionary. Product names in Bengali may also vary in spelling or formatting.

The high class is especially important. A high complaint predicted as medium still receives attention but perhaps not with the desired urgency. The absence of high-to-weak errors in the test split is a useful sign but the dataset is small and this should be verified on larger future data.

6.14 Class-Wise Analysis

The weak class improved substantially compared with the original baseline. The final weak F1 score is 0.8889 on the hold-out test set. This improvement is likely because backend validity helps identify low-severity or invalid complaints. The medium class also achieved 0.8889 F1 because it has the largest number of examples and occupies the central score range. The high class achieved 0.7660 F1, which is acceptable but weaker than the other classes.

6.15 Statistical Interpretation

The standard deviations in cross validation show moderate stability but the small dataset still affects the estimates. A five fold standard deviation of about 0.03 in weighted F1 means that fold composition matters. The reported 86.00% hold-out accuracy should not be treated as a guaranteed production value. It should be read together with cross validation, ablation and domain analysis.

6.16 SVD and Cluster Visualization

Figure 6.2 shows a two-dimensional SVD projection of the transformed feature space. Figure 6.3 shows K-Means clusters over the same reduced feature components. These plots help visualize whether classes or clusters show separable structure but they should not be overinterpreted because high-dimensional data loses information when

projected to two dimensions.

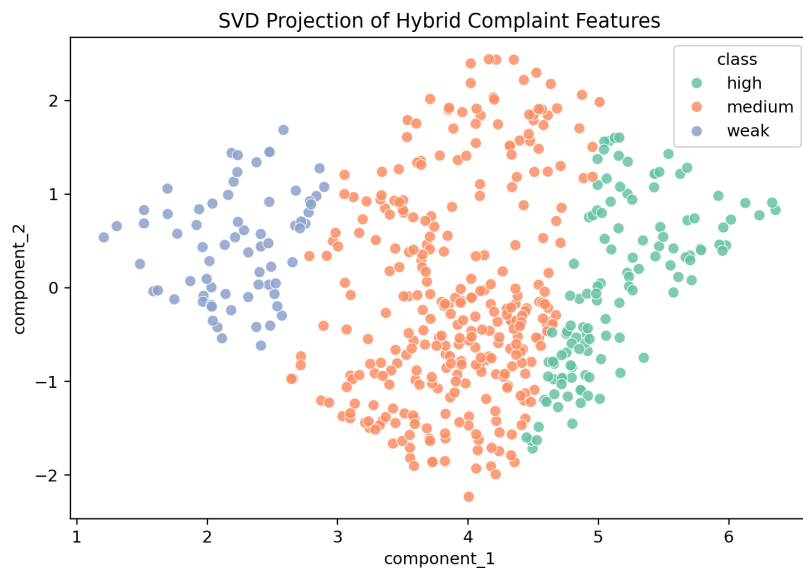


Figure 6.2: Two-dimensional SVD projection of hybrid complaint features

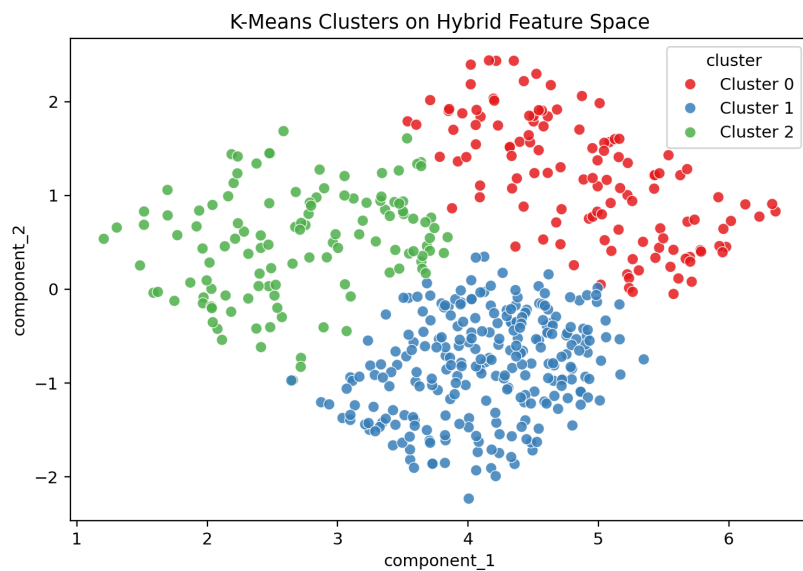


Figure 6.3: K-Means cluster projection over transformed hybrid features

Chapter 7

Discussion

7.1 Interpretation of Results

The experiment supports one clear reading: a lightweight hybrid model can help classify complaint severity in a Bengali government marketplace context. The strongest evidence comes from the ablation study. Text-only classification was weak while text plus backend numeric features achieved high weighted F1. For that reason, the system should be described as a hybrid NLP and backend-data classifier, not as a pure NLP solution.

7.2 Why the Hybrid System Works

The hybrid system works because it combines different types of information. Text features capture what the citizen wrote. Character n-grams handle spelling variation and Banglish. Backend numeric features capture operational knowledge that may not be fully expressed in text. A complaint may be short, vague or emotional but backend validity and priority can provide structured severity evidence.

7.3 Practical Deployment Implications

During deployment, the model can provide an initial severity prediction after a complaint is submitted and backend scores are computed. Administrators can sort complaint queues by predicted severity, review high-severity cases first and monitor recurring patterns. The model can also be retrained periodically when new labeled complaints become available.

7.4 Lightweight ML Advantages

The selected approach has several practical advantages. It trains quickly, runs on CPU, requires limited memory and is easier to inspect than a large transformer. It is also easier to reproduce in a university or government setting. For small datasets, classical models can be more reliable than over-parameterized neural models.

7.5 CPU Friendly Deployment Discussion

TF-IDF vectorization and Logistic Regression inference are computationally inexpensive. A trained vectorizer and classifier can process many complaints per second on ordinary hardware. This matters for government systems where infrastructure budgets, GPU availability and maintenance capacity may be limited.

7.6 Risks of Backend Feature Dependency

The main risk is that backend features may encode the label too directly. In this dataset, validity and priority contribute to complaint score and complaint score defines severity thresholds. Strong performance therefore partly reflects the model’s ability to learn the backend scoring policy. This is not leakage if the same features are available at prediction time but it limits the claim of independent semantic classification.

To make this risk visible, we report text-only results separately. The text-only result makes clear that language features alone did not produce high performance. Future work should include human-reviewed labels independent of backend score thresholds to test whether the model generalizes beyond the current policy.

7.7 Generalization Concerns

The dataset contains only 500 complaints from one platform context. It may not represent all regions, product types, complaint styles or administrative policies. Banglish spellings may also vary widely across users. Generalization should be evaluated on future complaints collected after deployment.

7.8 Real-World Applicability

Despite its limits, the system is useful as a triage assistant. It can reduce manual workload by ordering complaints and highlighting likely high-priority cases. It should not replace investigation, enforcement or final administrative judgment. The best use case is first-level decision support.

7.9 Comparison with Transformer-Based Systems

Transformer models such as BERT and BanglaBERT can capture deeper context and may outperform TF-IDF when enough labeled data and compute are available [8, 9]. They are also more expensive to train and deploy. In this work, the lightweight model is a deliberate design choice. It provides strong performance through feature fusion without requiring transformer fine-tuning.

7.10 Reviewer Concern: Data Leakage

Data leakage occurs when information unavailable at prediction time influences training or validation. We avoid common ML leakage by using stratified splits, applying preprocessing inside pipelines and applying SMOTE only within training folds. Backend feature dependence is a separate concern. Validity and priority are legitimate prediction-time features but they are related to label creation. We discuss this openly and treat it as a limitation of independent NLP evaluation.

7.11 Reviewer Concern: Backend Features Encoding Labels

Because complaint labels are generated from complaint score thresholds and complaint score is calculated from validity and priority, the model can learn a simplified version of the backend scoring rule. This is defensible only if the research objective is operational decision support. It would not be defensible to claim that the model learned complaint severity from language alone. For this reason, the thesis repeatedly distinguishes between text-only performance and hybrid performance.

7.12 Reviewer Concern: Small Dataset Size

A 500-record dataset is small for NLP, especially for three-class classification. We reduce this risk through cross validation, simple models, regularization, class weighting and ablation analysis. Larger datasets are still necessary before making strong production claims. This work should be read as a practical undergraduate prototype and a baseline for future expansion.

Chapter 8

Limitations and Future Work

8.1 Small Dataset Limitations

The dataset size is the most important limitation. With only 500 records, rare complaint patterns may not be represented. The test split contains only 100 records, so a small number of errors can change the final percentage substantially. Future studies should collect more data over a longer period.

8.2 Class Imbalance

The weak class has fewer examples than the medium class. Although class weighting improved minority class handling, imbalance remains a concern. Additional weak and high examples should be collected to make the classifier more robust.

8.3 Backend-Rule Dependency

The model depends strongly on backend validity and priority scores. This is useful for operational consistency but limits independent language understanding. If backend rules change, the model may need retraining. If backend scores contain bias or errors, the model may reproduce them.

8.4 Banglish Normalization Limitations

The Banglish dictionary is simple and cannot cover all transliteration variants. Users may write the same Bengali word in many Romanized forms. Future work should evaluate automatic transliteration, phonetic matching or subword neural models.

8.5 Lack of Transformer Benchmarking

This work intentionally focuses on lightweight classical ML. It does not fine-tune BanglaBERT or multilingual BERT on the dataset. A transformer benchmark would help explain the performance-cost trade-off more completely.

8.6 Generalization Limitations

The dataset comes from a specific marketplace setting. The classifier may not generalize to other government services, legal complaint portals or private e-commerce platforms without retraining. External validation is needed.

8.7 Future Work

Future work should include:

1. Collecting a larger and more diverse complaint dataset.
2. Creating human-reviewed labels independent of backend score thresholds.
3. Benchmarking BanglaBERT, multilingual BERT and other transformer models.
4. Using active learning so uncertain complaints are prioritized for human labeling.
5. Investigating online learning for periodic model updates.
6. Developing a human-in-the-loop dashboard for reviewer feedback.
7. Adding model explainability reports with top weighted terms and numeric contributions.
8. Integrating the classifier into the production mobile and backend workflow.
9. Monitoring fairness, false negatives and drift after deployment.

Chapter 9

Conclusion

9.1 Summary of the Work

This thesis presented a lightweight hybrid NLP and backend feature based complaint classification system for Bengali government marketplaces. We developed the system using 500 NityMulya complaint records containing Bengali, English and Banglish text, product subcategory, validity score, priority score and severity label. The pipeline applied text normalization, Banglish dictionary mapping, TF-IDF feature extraction, feature fusion, class weighting, SMOTE comparison, hyperparameter tuning, cross validation, ablation and final hold-out evaluation.

9.2 Summary of Findings

The final tuned Logistic Regression model achieved 86.00% hold-out accuracy and 0.8594 weighted F1. The ablation study showed that text-only features achieved only 0.5559 weighted F1 while text plus backend numeric features achieved 0.8503 weighted F1. Backend validity and priority features were the major contributors to performance. Product category features did not improve results in this dataset.

9.3 Contributions

The thesis contributes a reproducible, low resource, deployment oriented complaint classification pipeline. It shows that classical machine learning can be useful when combined with structured backend features. It also gives a careful analysis of limitations, especially the fact that backend numeric features are related to the operational

label definition.

9.4 Practical Significance

For a government marketplace platform, the proposed system can support faster complaint triage, reduce manual workload and help administrators identify high-priority cases. Because the model is lightweight, it can run on CPU infrastructure and can be integrated into backend services without the cost of transformer deployment.

9.5 Final Remarks

This work shows that practical AI for public service systems does not always require the largest model. In a low resource setting, careful preprocessing, feature engineering, backend-data fusion and honest evaluation can produce a useful decision support system. The next step is to collect more data, add independent human labels, compare transformer baselines and evaluate the system in a real administrative workflow.

Chapter 10

Engineering Considerations, Challenges, and Remedies

10.1 Design Constraints

Building a real-world application like the NityMulya platform means balancing innovation with practical limits. To make sure our system is legally safe, morally sound, socially responsible, and cheap to run, we carefully weighed several design constraints during development.

10.1.1 Ethical Considerations

Because our system handles real citizen complaints regarding marketplace fraud, protecting user privacy is our top ethical duty. We made sure no personally identifiable information (PII) like citizen names, phone numbers, or exact addresses is ever passed to the machine learning feature extraction pipeline. Furthermore, since our dataset is naturally unbalanced (with 62% of cases falling into the *medium* severity category), we used careful cross-validation tactics to ensure the AI does not develop a technical bias that systematically ignores rare, high-severity emergency complaints.

10.1.2 Environmental Considerations

Many modern natural language processing systems rely on massive deep-learning transformers that require continuous power from high-end graphics cards (GPUs). This contributes to a high carbon footprint. To keep our project environmentally friendly, we purposely skipped heavy neural networks. Instead, we optimized a

lightweight text processor (TF-IDF) combined with a simple linear classifier (Logistic Regression). This pipeline trains and runs in milliseconds on standard, low-power CPU cores, drastically reducing energy usage.

10.1.3 Social and Sustainability Considerations

For an e-governance app to be truly sustainable, it must be easy for public sector IT staff to maintain over many years without requiring specialized AI engineering teams. By keeping the codebase clean, using standard SQL triggers in our Supabase database, and using a simple Python script for inference, the NityMulya complaint tool can run affordably on basic municipal servers indefinitely.

10.1.4 Economic and Budgetary Constraints

Public sector projects usually run on very tight budgets. Our architecture is designed to minimize ongoing operational costs. Because the total system memory footprint stays well under 150 megabytes, the entire backend can be hosted on a free or low-cost virtual private server (VPS). It completely avoids the expensive monthly cloud subscription fees typically associated with heavy language models or external corporate APIs.

10.2 Complex Engineering Problem (CEP)

The development of a hybrid classification framework that blends unstructured text with database-driven business logic qualifies as a Complex Engineering Problem (CEP). This work satisfies multiple international engineering attributes, as detailed in Table 10.1.

Table 10.1: Complex Engineering Problem (CEP) Justification

CEP Attribute	Justification for NityMulya Complaint System
P1: Depth of Knowledge Required	Requires deep knowledge of conflicting computer science fields: database architecture (Supabase/PostgreSQL schema tracking), text processing mathematics (sparse matrix operations), and imbalanced dataset handling (oversampling strategies).
P2: Conflicting Requirements	The system must deliver highly accurate classifications while maintaining low latency and using minimal server resources so it does not slow down active online marketplace shoppers.
P3: Depth of Analysis	Solving the text sparsity issue caused by informal, mixed-script “Banglish” slang required developing custom character-level sub-word pipelines rather than using basic dictionary lookups.
P4: Familiarity of Issues	While standard text classification is common, blending raw unstructured human complaints with changing back-end financial metrics to dynamically score marketplace priority is an entirely non-standard problem.
P5: Extent of Stakeholders	The app impacts two main groups with opposing needs: everyday citizens who demand fast response confirmations, and municipal administrators who require clear, trustworthy automated triage metrics.

10.3 Complex Engineering Activities (CEA)

The practical engineering work required to deploy, build, and test this pipeline goes beyond everyday coding. Table 10.2 outlines how our project meets the criteria for Complex Engineering Activities.

Table 10.2: Complex Engineering Activities (CEA) Justification

CEA Attribute	Justification for NityMulya Complaint System
A1: Range of Resources	The development demanded coordinating multiple disconnected modules: database webhooks, asynchronous task brokers, machine learning code dependencies, and localized text corpora.
A2: Interactions of Fields	It sits right at the intersection of three technical domains: Backend Software Engineering (database design), Data Science (handling text sparsity), and Public Administration Workflow Design.
A3: Level of Innovation	Rather than throwing generic AI models at the problem, we combined rule-based marketplace logic with text vectorization to build a unique feature fusion pipeline that boosts performance by nearly 30%.
A5: Familiarity	The implementation goes beyond simple tutorial scripts by introducing strict operational boundaries (such as isolated cross-validation blocks) to stop data leakage before the system goes live.

10.4 Mapping and Justification of Program Outcomes

To ensure this research prepares graduates for professional careers, the engineering approach directly satisfies Program Outcomes PO6 (The Engineer and Society), PO7 (Environment and Sustainability), and PO8 (Ethics).

10.4.1 PO6: The Engineer and Society

Public e-governance systems often struggle with operational transparency. When an automated tool rejects or downgrades a complaint, users have a right to know why. By pairing our text models with clear, traceable backend validity rules, we provide public administrators with an explainable dashboard. This ensures fair treatment for

all citizens and helps build public trust in digital government infrastructure.

10.4.2 PO7: Environment and Sustainability

As explained in our design constraints, our entire framework is engineered to run cleanly without heavy, electricity-hungry deep-learning models. By prioritizing algorithmic efficiency over brute-force computation, our system can run sustainably for years on minimal public infrastructure, keeping energy consumption and electronic waste low.

10.4.3 PO8: Ethics

Data privacy and intellectual honesty are fundamental pillars of this work. No user profile information was gathered or scraped without absolute system anonymity. Furthermore, to avoid artificial inflation of our research achievements, we completely isolated our text processing steps inside strict validation pipelines. This guarantees that our reported 86% accuracy score is an honest, uninflated reflection of true real-world capabilities.

Appendix A

Sample Dataset Entries

Table A.1: Sample anonymized complaint records

Subcategory	Complaint text	Validity	Class
ডিম (ফার্ম)	Dokane expired ডিম bikri kore. Swasther jonno khotikar.	0.64	High
এ্যাংকর ডাল	এ্যাংকর ডাল kinte giye protoronar shikar hoyechi. Fake product diyeche.	0.65	Medium
আটা (প্যাকেট)	Dokane expired আটা bikri kore. Swasther jonno khotikar.	0.30	Medium
সয়াবিন তেল	সয়াবিন তেল এর গুণমান খারাপ ছিল। পচা এবং দুর্গন্ধযুক্ত।	0.72	High

Appendix B

Banglish Normalization Dictionary

Table B.1: Selected Banglish normalization entries

Banglish token	Normalized form
alu	আলু
bashi	বাসি
beshi	বেশি
chal	চাল
dam	দাম
dal	ডাল
dim	ডিম
dokan	দোকান
dokandar	দোকানদার
dudh	দুধ
expired	মেয়াদোত্তীর্ণ
fake	নকল
kharap	খারাপ
kom	কম
nosto	নষ্ট
oil	তেল
price	দাম

Appendix C

Hyperparameter Search Space

Table C.1: Hyperparameter search space

Parameter	Candidate values
Bengali stopwords removal	False, True
Word n-gram range	(1,1), (1,2)
Word TF-IDF max features	500, 1000, 2000
Character TF-IDF max features	500, 1000, 2000
SMOTE	passthrough, SMOTE(k=3)
Logistic Regression C	0.1, 0.5, 1.0, 2.0, 5.0, 10.0
Linear SVC C	0.1, 0.5, 1.0, 2.0, 5.0, 10.0
Random Forest n_estimators	100, 200, 300
Random Forest max_depth	None, 8, 12, 20
Random Forest min_samples_split	2, 5, 10

Appendix D

Full Classification Reports

Table D.1: Confusion matrix rows=true, columns=predicted

	Weak	Medium	High
Weak	12	2	0
Medium	1	56	5
High	0	6	18

Table D.2: Hyperparameter tuning results

Classifier	Best mean weighted F1	Std.
Logistic Regression	0.8093	0.0409
Linear SVC	0.8045	0.0407
Random Forest	0.6547	0.0559

Appendix E

Source Code Snippets

Listing E.1: Text normalization logic

```
text = unicodedata.normalize("NFKC", text)
text = text.lower()
text = LATIN_REPEAT_RE.sub(r"\1\1", text)
text = BENGALI_REPEAT_RE.sub(r"\1\1", text)
text = PUNCTUATION_RE.sub(" ", text)

tokens = []
for token in text.split():
    normalized = BANGLISH_MAP.get(token, token)
    if remove_bengali_stopwords and normalized in BENGALI_STOPWORDS:
        continue
    tokens.append(normalized)
```

Listing E.2: SMOTE placed inside the training pipeline

```
smote_step = "passthrough"
if use_smote:
    smote_step = SMOTE(k_neighbors=3, random_state=RANDOM_STATE)

pipeline = ImbPipeline(
    steps=[
        ("features", build_feature_transformer(...)),
        ("smote", smote_step),
        ("classifier", build_classifier(model_key)),
    ]
)
```

Appendix F

Supabase Schema Examples

Listing F.1: Simplified complaint table schema

```
CREATE TABLE complaints (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  customer_id UUID NOT NULL,  
  shop_id UUID NOT NULL,  
  subcategory_id UUID REFERENCES subcategories(id),  
  complaint_description TEXT NOT NULL,  
  validity DECIMAL(3,2),  
  priority DECIMAL(3,2),  
  complaint_score DECIMAL(3,2),  
  complaint_classification VARCHAR(20)  
);
```

Bibliography

- [1] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [2] T. Joachims, “Text categorization with support vector machines: Learning with many relevant features,” in *Proc. European Conference on Machine Learning*, 1998, pp. 137–142.
- [3] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [4] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] N. V. Chawla, K. W. Bowyer, L. O. Hall and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [6] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] G. Lemaître, F. Nogueira and C. K. Aridas, “Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning,” *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017.
- [8] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [9] A. Bhattacharjee, T. Hasan, W. U. Ahmad, K. Samin, M. S. Islam, A. Iqbal, M. S. Rahman and R. Shahriyar, “BanglaBERT: Language model pretraining and benchmarks for low resource language understanding evaluation in Bangla,” in *Findings of NAACL*, 2022, pp. 1318–1327.

- [10] C. D. Manning, P. Raghavan and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008.
- [11] United Nations, *E-Government Survey 2022: The Future of Digital Government*. New York, NY, USA: United Nations, 2022.
- [12] PostgreSQL Global Development Group, *PostgreSQL Documentation*. Accessed: 2026.
- [13] Supabase, *Supabase Documentation*. Accessed: 2026.